

AA/EE/ME 548

Linear Multivariable Control

Spring 2026

Course website: <https://uw-ctrl.github.io/lmc-book/>

Bookmark this

Key supporting infrastructure

- **Zoom recordings** – See Canvas
- **Submitting deliverables** – See Canvas
- **Asking questions related to course material/concepts** – Use Ed Discussion
- **Reaching out to teaching team regarding personal matters** – Email aa548-spr26-staff@uw.edu
- **Lecture content and information** – See course website <https://uw-ctrl.github.io/lmc-book/>

Coding in this class

Homework and course project will require you to code in Python. In particular, use JAX.

Why?

- A large part of controls is designing and testing out the behavior
- Modern and state-of-the-art software packages in Python make the translation from theory to practice very accessible, simple, and efficient

What is this course about?



<https://www.youtube.com/watch?v=joIsgP9StAY>



Robotics Today

A Series of Technical Talks

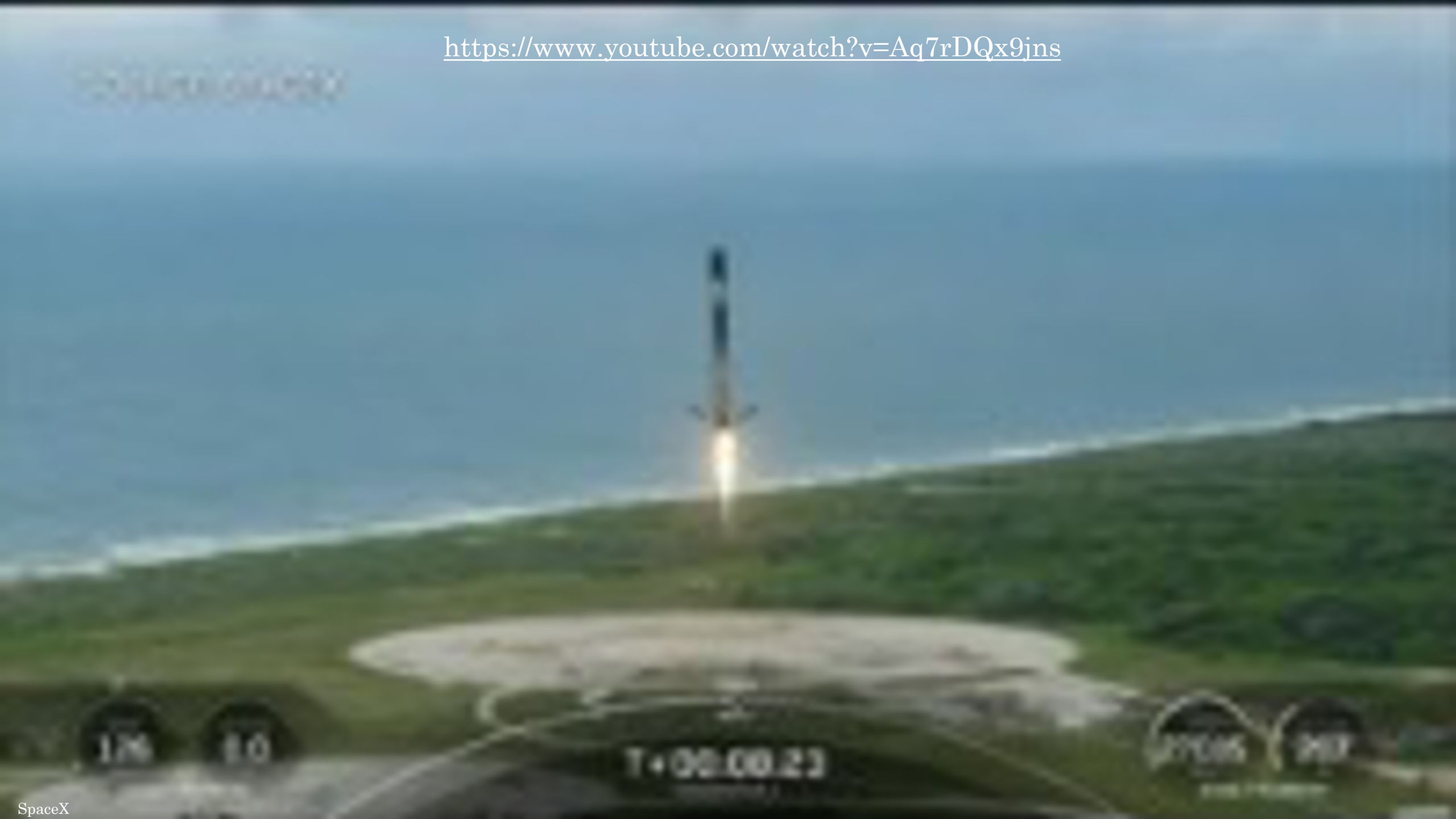
Scott Kuindersma

Boston Dynamics

***Recent Progress on Atlas,
the World's Most Dynamic
Humanoid Robot***

June 28, 2020

<https://www.youtube.com/watch?v=Aq7rDQx9jns>



17%

8.0

T+00:00.23

27005

2407

FALCON HEAVY

What is the secret sauce?

Exploiting knowledge about the dynamics and applying optimization-based control

MODEL-PREDICTIVE CONTROL

Having identified the boxes, ramps, or barriers in front of the robot and planned a sequence of maneuvers to get over them, the remaining challenge is filling in all of the details needed for the robot to reliably carry out the plan.

Atlas' controller is what's known as a *model-predictive controller* (MPC) because

VI. I

To transform quickly, the dynamic system. While not required to produce a useful modified trajectory, the nearer this nominal trajectory is to being feasible, the better since less approximation error will be introduced by the linearization. The method of discretization is chosen to limit the resulting error as much as possible.

A. Linearization

To include the vehicle dynamics as equality constraints in a quadratic program (QP), we place the equations of motion in the following linear form:

$$\dot{x}(s) = A(s)x(s) + B(s)u(s),$$

where $x(s) = [\Delta t \ e \ \Delta V \ \sigma]^T$, $u(s) = [\Delta a_x \ \Delta a_y]^T$; x notes a perturbation from the nominal value at that point.

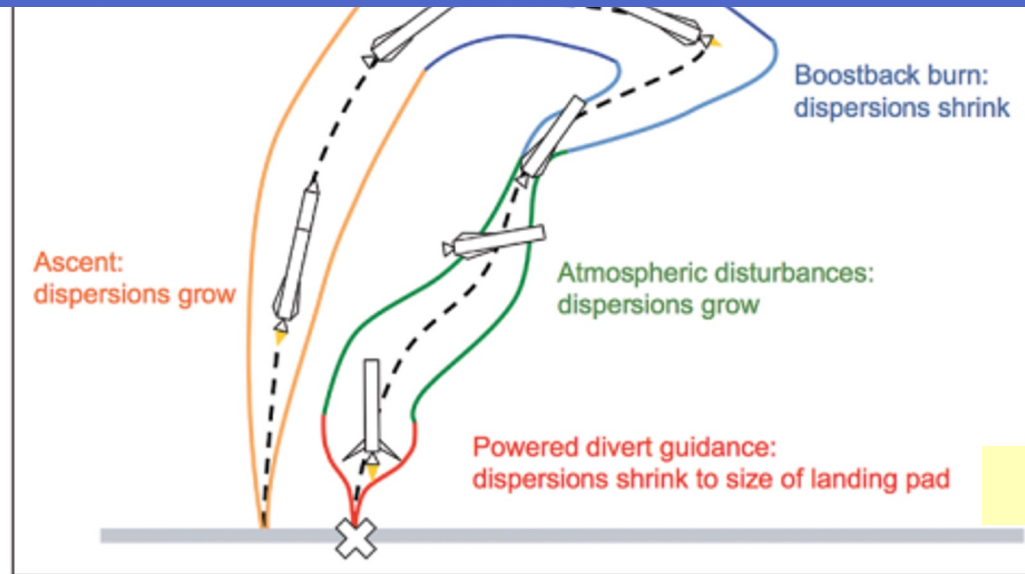


FIGURE 4 Phases of an F9R return-to-launch-site mission. The colored lines represent the largest possible variations in the trajectory, known as dispersions.

Earth's atmosphere is 100 times as dense as that of Mars, aerodynamic forces become the primary concern rather than a disturbance so small that it can be neglected in the trajectory planning phase. As a result, Earth landing is a very different problem, but SpaceX and Blue Origin have shown that this too can be solved. SpaceX uses CVXGEN (Matingley and Boyd 2012) to generate customized flight code, which enables very high-speed onboard convex optimization.

momentum as it moves through the course.

The ultimate goal:

Control input to execute

Observations on the world and system

$$u = \pi(o)$$

Controller that encodes desired
behavior and constraints

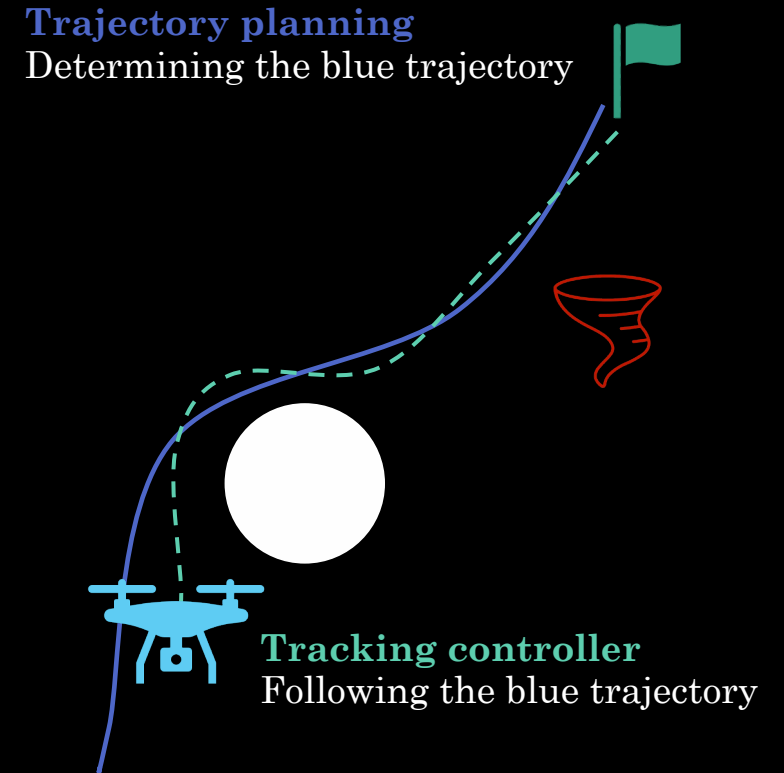
This class: $u = \pi(x)$
State

Closed-loop control for real-world systems is hard...

- In this course...
 - Explore how we design controllers by approximating real-world (i.e., nonlinear) systems as *linear* systems
 - Cover a number of control design techniques & show that they are tractable for **linear systems**
 - Focus on *optimization-based* control (very powerful thanks to modern computing)
 - Connect to some modern learning-based methods (new this year)
- Emphasis on theory (intuition) and implementation

Scope of this course

- A model-based approach to control synthesis
 - Assume we have a reasonably good model of the system
- What you can after taking this class
 - Frame a control problem as an optimization problem
 - Synthesize a trajectory given goals and constraints, i.e., go from point A towards point B while avoiding things + additional constraints (computing the nominal trajectory)
 - Construct tracking controllers, i.e., system to follow a nominal trajectory



Potential applications

- Human-robot interactions
- Spacecraft landing/docking
- Aircraft landing/flying
- Delivery robots
- Autonomous driving

Outside the scope of this course

- What this course is **not** about
 - Model-free approaches (no model), i.e., reinforcement learning or deep learning
 - But can use learned components for some parts
 - Optimization theory
 - We will cover some basics, but we will not go into depth
 - State-estimation/filtering course
 - Programming class
- Topics lightly touched but beyond the scope of this course
 - Stochastic control, robust control, nonlinear control, adaptive control, state estimation, reinforcement learning